Rebuild around the project

Study findings for a construction ERP covering tendering through handover and snagging.

The strongest confirmed problem is the organisation and continuity of work. The ERP contains substantial construction functionality, but users must navigate module names, transaction lists, generic project forms, and several dashboard layers to reach it.

A new application should make the project, work package, location, current owner, and next action clear throughout the lifecycle. Rebuilding the same menu structure in new technology would preserve much of the present difficulty.

What the live walkthrough established

Confirmed finding	Implication for the rebuild
42 primary administrator entries; 267 leaf menu controls	Reduce primary choices and organise daily work by role and project context.
Project entry opens on a map; key links are inside More	Make current work and exceptions the project landing content.
Requisition, PO, and receipt are linked, but context fades	Keep a visible project / package / location context across separate records.
A flat 302 snag is structurally located only at Main site	Validate floor, unit, zone, and system relationships for tower operations.
Variation and closeout samples focus on commercial value	Validate instruction/time linkage and technical handover separately.

What is already useful

Role filtering, linked records, current/superseded drawing revisions, textual statuses, cumulative measurements, and approval history are visible. These business capabilities should survive the simplification.

Boundary: this is an expert read-only UI walkthrough and a proposed rebuild brief. It does not prove full functional coverage, accounting correctness, accessibility compliance, security, performance, or usability with actual staff. No application code was started.

Expanded study: your twelve requested areas

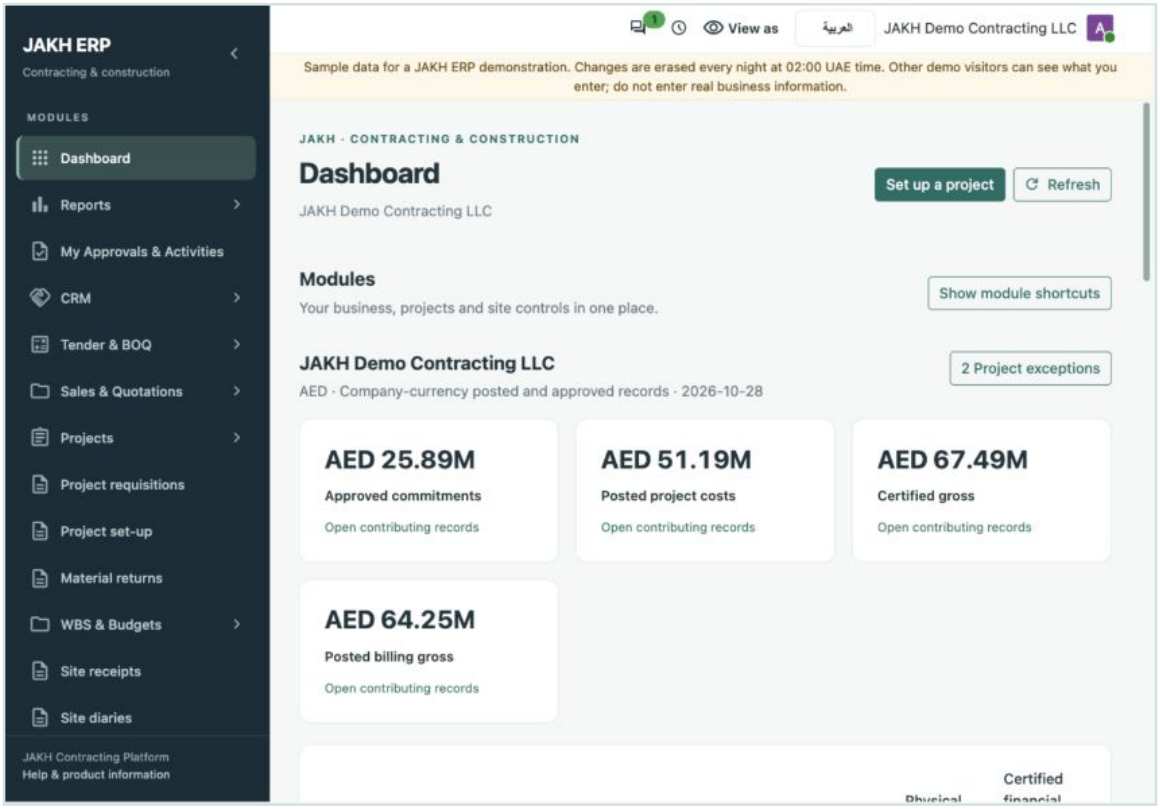
Pages **11-12** map existence and need. Pages **13-26** examine each area, with extra cost definitions and an item example on pages **18-19**. Pages **27-30** connect the findings to a simpler structure, original-code scope and validation scenarios. Evidence and limits are on pages **31-32**.

The follow-up confirms NCR records, asset/maintenance records, linked subcontract finance and granular cost-report foundations. Full waste, delay, item-level cash and cross-workflow controls still need depth validation. An inspected company cash report is blocked by a missing selected policy.

Navigation needs a new structure

Evidence: E39 administrator dashboard, E41 expanded navigation inventory, E30/E50/E53 role dashboards.

Role	Primary entries	What this proves
Administrator	42	Broad modules and individual transactions share one level.
Project Manager	20	Role filtering still leaves a long operational menu.
Site Engineer	10	A smaller role-specific menu already exists.
Estimator	9	Tender work can be presented with fewer primary areas.



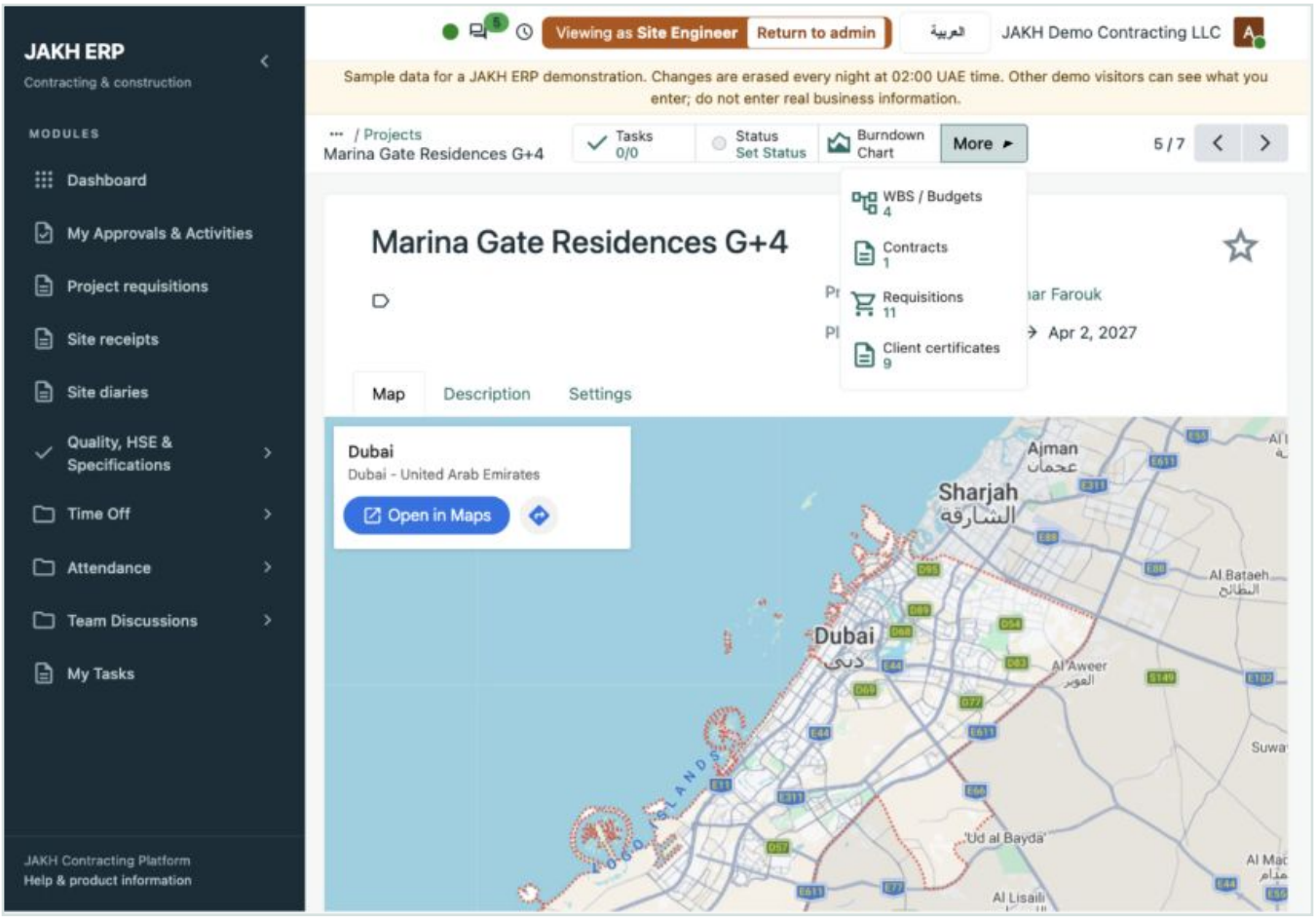
E39. Administrator in JAKH Demo Contracting LLC. The screenshot shows the first portion of the scrollable sidebar.

The fully expanded administrator tree contains **363 button controls: 96 groups and 267 leaf buttons**, with paths reaching four menu levels. Leaf counts are navigation controls, not a count of unique features.

Repeated leaf labels include Products (6), Customers (4), Settings (12), Dashboard (4), Payments (3), and My Tasks (2). Their targets were not all compared; identical labels do not prove duplicate records or duplicate functionality.

A project should open on work

Evidence: E32 project entry and E33 its More menu, viewed as Site Engineer.



E33. Marina Gate Residences G+4 opens on Map / Description / Settings. WBS, Contracts, Requisitions and Client certificates sit under More.

Observed friction

The main project view prioritises a large Dubai map, generic task counts, and project metadata. It does not immediately answer what is happening on site, what is late, what needs approval, or which floor/package needs attention.

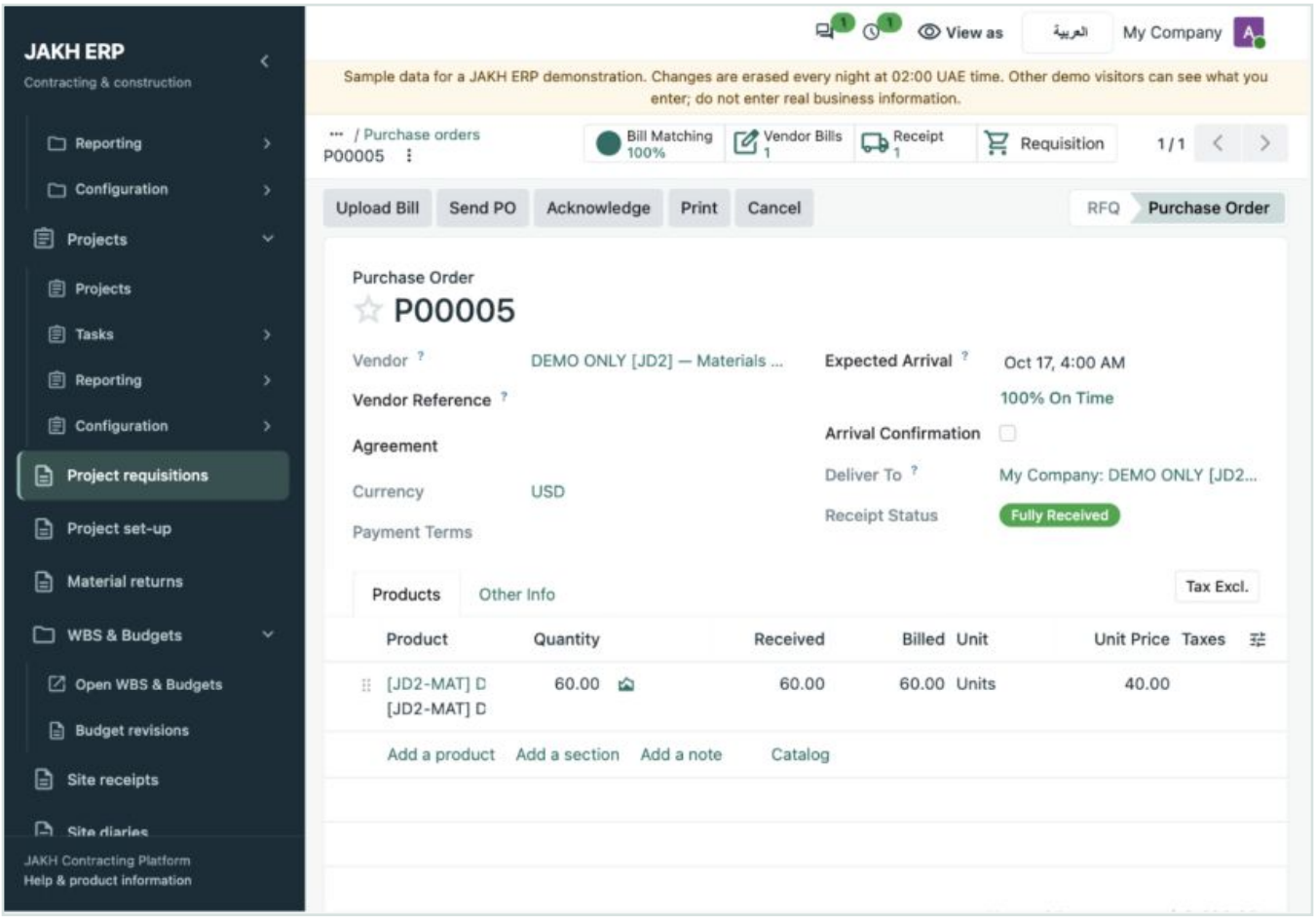
Requirement to validate

Use a project workspace with a stable identity and location/package filter. Put current actions, blockers, recent progress, and relevant documents within that context. Treat maps and administrative settings as supporting information.

Waiting records should name the next owner and expected review date, including when the viewing user cannot approve. The absence of an approval button for a site role may be correct permission behaviour.

Keep the procurement chain connected

Observed path: requisition -> linked PO list -> purchase order -> completed receipt. Evidence E19-E22.



E21. P00005 shows received/billed quantities and links to the receipt, bill, and requisition. Its main view does not give project/WBS/location the prominence seen on the requisition.

What works, and where context weakens

The requisition includes the project, WBS, required date, planning-demand reference, and requested location. The PO exposes the vendor, arrival, receipt status and quantities. The receipt exposes the source PO and stock movement. The linkage is present; the user must reconstruct its construction context.

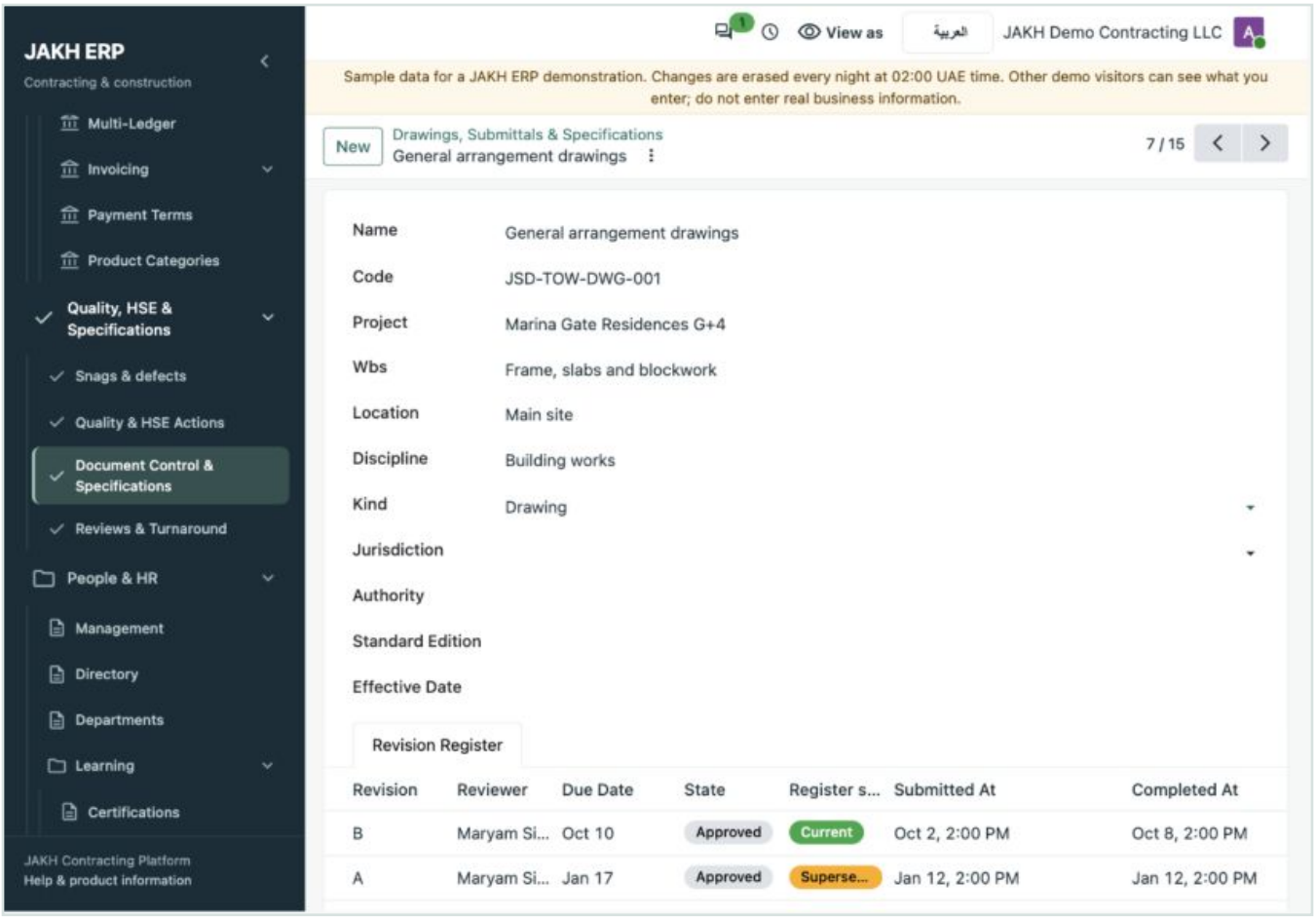
Requirement to validate

Carry project, package and location visibly across demand, request, quotation, order, receipt, inspection, issue, installation, and invoice. Link distinct controlled records rather than merging them into one editable document.

In this sample, a link labelled 1 Purchase orders opens an intermediate single-row list before the PO. Test direct opening when there is exactly one related record. Partial delivery, rejection, returns and billing exceptions were not executed.

Technical controls exist

Evidence: E43 document register, E44 revision history, E45 document-category dropdown.



E44. General arrangement drawing JSD-TOW-DWG-001 shows approved revision B as Current and approved revision A as Superseded.

Preserve revision meaning

The current-versus-superseded distinction is useful. A register also needs to let a site user find the correct approved document for the floor, zone, trade and activity without searching through unrelated projects.

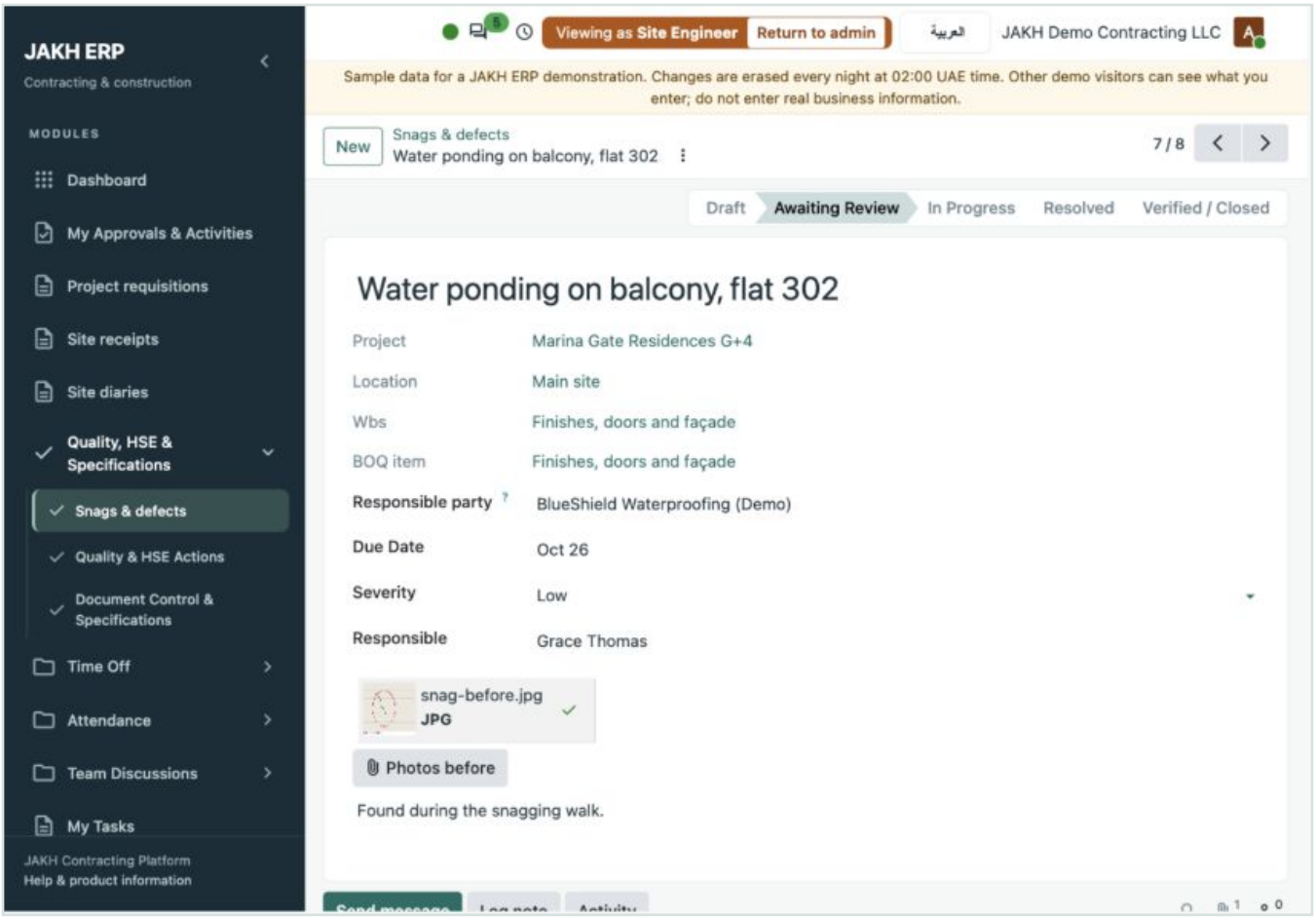
The category control offers Drawing, Submittal, Specification, Method Statement, Inspection / Test Plan, and Correspondence. The fuller register contained 15 drawing records; other category workflows were not demonstrated by populated samples.

Still to validate

Dedicated RFI routing, drawing release/transmittal, consultant response codes, inspection hold points, document distribution and external-party review. A category called Correspondence does not by itself demonstrate an RFI workflow.

Tower work needs structured locations

Evidence: E34 snag register, E35 snag detail, E37-E38 submitted site diary.



E35. Water ponding on balcony, flat 302 is linked to a project, WBS, BOQ item, contractor and person. Its structured Location is Main site.

A concrete retrieval problem

The eight displayed open snags all use Main site, including titles referring to a flat or individual villa. For these records, finding work by floor or unit depends on free text. This is a sample-data limitation and an information-structure risk; it does not prove the application cannot represent locations.

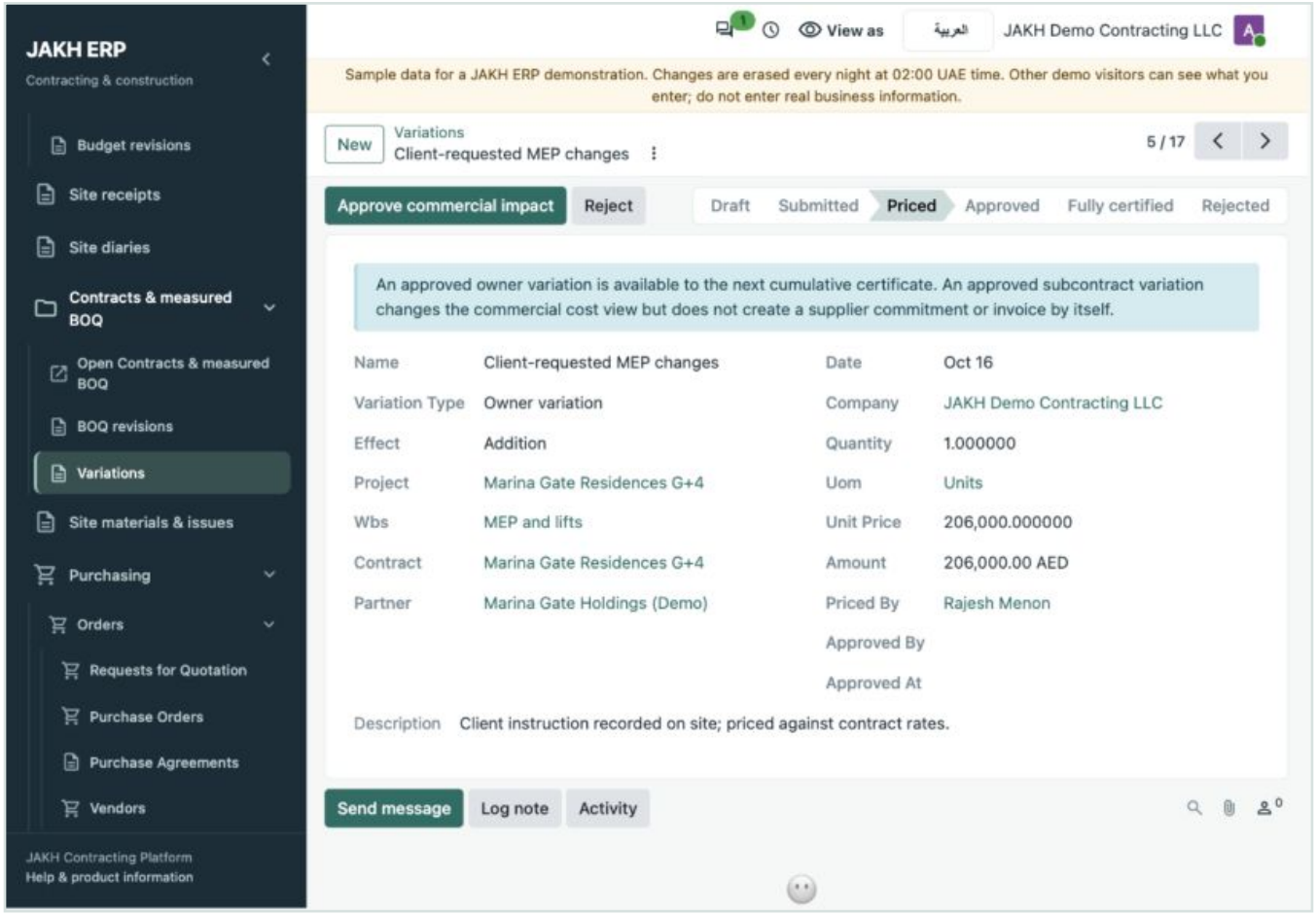
Requirement to validate

Model project -> tower/building -> floor -> unit/zone, with systems and packages linked separately. Test bulk snagging, photos, contractor assignment, rectification, reinspection, reopening and partial handover by location.

The site diary has purposeful Manpower, Equipment, Photos, and Work and events tabs. Keep useful grouping. Its Submitted state and the snag's Awaiting Review state need clearer next-review ownership in the captured views.

Connect changes to their consequences

Evidence: E46-E47 variation records; E15 client certificate and E29 subcontract service entry.



E47. A priced MEP variation has project/contract links, quantity, rate, value, pricing owner, and approval actions. No instruction link, notice deadline, or time impact is visible on this record.

Commercial coverage is present

The sampled client certificate contains cumulative quantities, prior/current/remaining measurements, approved variations, retention/recovery values, an invoice link, and technical/commercial/finance review history. A submitted subcontract service entry also has package, location and quantity context.

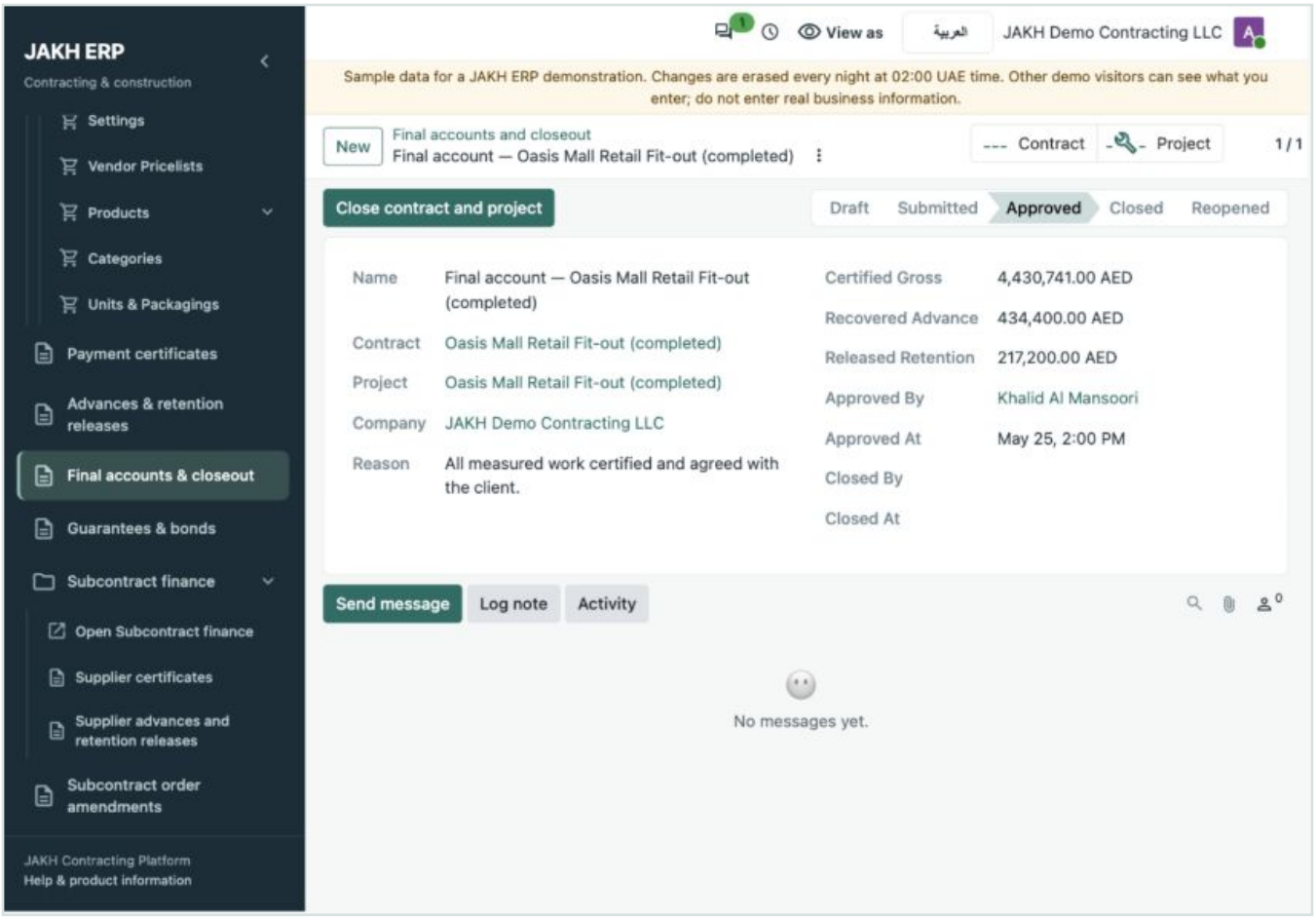
The chain to validate

Source instruction -> required notice -> estimate and cost/time impact -> review -> approval or dispute -> budget/programme change -> measured work -> certification -> invoice -> cash. Include work proceeding before approval and rejected or partially certified amounts.

Missing fields on the sampled variation are not proof of system-wide absence. Contract-specific notice rules, approval authorities and finance treatment require validation with your business owners.

Handover is more than a final account

Evidence: E48-E49 final-account workflow; E34-E35 snagging. No closure action was executed.



E49. The completed retail-fit-out project has an Approved final account and a Close contract and project action. The visible detail focuses on certification, advance recovery and retention.

Separate three completion questions

Technical readiness: are inspections, testing/commissioning, authority requirements, as-builts, O&M manuals, warranties, training and required snags complete?

Client acceptance: which building, floor, unit or system is accepted, by whom, with what outstanding obligations and defect-liability dates?

Commercial settlement: are measurements, variations, claims, subcontract settlements, retention and final balances agreed?

The sampled final-account screen does not visibly present a structured technical-readiness pack. Backend closure checks and wider handover workflows were not tested. Define the three tracks explicitly before designing the replacement.

Construction coverage map - 1

Observed means visible in this run. Not demonstrated means unverified, not proven absent. Execution/approval transitions were not performed.

Lifecycle area	Current evidence	Requirement / next verification
Tender pipeline / bid decision	Observed: generic CRM entry; estimator preview has no matching opportunities (E51).	Validate bid/no-bid, tender deadlines, qualifications, clarifications, tender security and opportunity access.
BOQ / estimate / revisions	Observed: estimates, study versions, pricing, review and transfer links (E05, E52).	Validate take-off detail, quote comparison, exclusions, alternate bids, revision comparison and approval limits.
Award / project mobilisation	Observed: estimate-to-quotation/project transfer; setup entry (E05-E07, E26).	Validate contract obligations, responsibilities, programme and budget baseline; direct-award path.
Project / tower locations	Observed: location records; fuller building sample uses Main site (E35, E44).	Validate tower/floor/unit/zone hierarchy and large-volume navigation. G+4 is not a high-rise scale test.
Budget / cost codes / forecasts	Observed: BOQ/location cost report, source links and approved cost-to-complete forecasts (E66-E69, E90).	Validate item/resource detail, revisions, commitments, incurred costs and cash reconciliation. See pages 17-19.
Programme / look-ahead	Observed: 145 progress records; the detail says P6 is schedule master (E79-E80).	Validate activity logic, baselines, imports, look-ahead and delay/EOT linkage. See page 22.
Drawings / submittals / documents	Observed: current/superseded drawings and six categories (E43-E45).	Validate submissions, transmittals, distribution, consultant response and overdue review ownership.
RFIs / instructions	Not demonstrated as a dedicated workflow in the inspected views.	Trace RFI -> response -> drawing/work instruction -> cost/time consequences. Do not treat a label as proof.
Materials / procurement	Observed: demand/request/order/receipt links (E19-E22).	Validate quote comparison, approvals, lead times, partial/rejected delivery, returns and material inspection.
Stores / site issues / installation	Navigation and demand links observed; full operational chain not walked.	Trace stock source, reservation, transfer, issue, installed quantity, wastage and cost allocation.
Subcontract award / amendments	Navigation and subcontract order/service links observed (E28-E29, E41).	Validate package scope, exclusions, measurement rules, changes, commitments and subcontract securities.
Subcontract measurement / payment	Observed: service entries, supplier certificate, bill and retention agreement links (E29, E70-E72).	Validate application -> acceptance -> settlement; rejected work, deductions and correction. See page 16.

Construction coverage map - 2

This map is a requirements register for the new ERP. Supporting functions remain in scope even when sampled only through navigation.

Lifecycle area	Current evidence	Requirement / next verification
Daily reports / labour / plant	Observed: populated diaries with resources and work/events (E36-E38); allocation/feed entries.	Validate who submits/approves, location/package detail, hours, equipment use and cost-feed reconciliation.
Quality / HSE / inspections	Observed: non-conformances, inspections and risk assessments; NCR resolution/lifecycle (E56-E59).	Validate ITP hold points, cause/corrective action, independent verification and safety controls. See page 13.
Variations / notices / delay	Observed: price/approval states; time/notice chain not visible in sample (E46-E47).	Define instruction evidence, notice dates, cost/time assessment, disputed work and linked approval history.
Client measurement / certification	Observed: cumulative BOQ/variation measurement and staged approval (E15).	Validate claimed vs certified/rejected values, evidence, corrections, prior quantities and invoice progression.
Finance / cash / accounting	Observed: bill and matched receipt sources. Cash report blocked by policy selection; visible policy list empty (E70, E74-E76, E91-E92).	Validate forecasts, policy setup, reconciliation and original posting/localisation. See pages 15 and 18.
Approvals / delegation / exceptions	Observed: role routes/history; PM activity queue has two threshold tasks (E15-E16, E54).	Verify queue coverage for all pending documents; delegation, absence cover, rejection/resubmission and independence.
Testing / commissioning	Not demonstrated as an end-to-end workflow in inspected records.	Define system/asset tests, dependencies, witnesses, results, failures/retests and readiness evidence.
Snagging / rectification	Observed: populated snags, owners, before photo and status sequence (E34-E35).	Validate after-photo requirements, independent verification, reopening, bulk capture and unit/system filtering.
Handover / defect liability	Not demonstrated as a structured handover pack in the sampled final-account view (E49).	Define partial acceptance, as-builts, O&M, warranties, training, DLP dates, outstanding work and responsibility.
Final accounts / retention	Observed: financial final-account workflow; sample Approved and Closed records (E13, E49).	Validate technical/contractual readiness gates, settlement, retention release and auditable reopening.
People / assets / shared masters	Observed: equipment, serial/cost and maintenance detail (E61-E64). Most HR/fleet scope remains navigation-only.	Validate asset custody, utilisation, hire, finance and handover; retain required HR/shared scope. See page 14.
Portals / integrations / migration	Role chooser exposes client/subcontractor portals; schedule and cost feeds have entries.	Verify portal permissions, external reviews, integration ownership, data/attachment export, migration and cutover.

Your 12 areas: existence and need

Expanded scope map 1 of 2. Follow-up inspection: administrator, JAKH Demo Contracting LLC, AED, 28 October 2026.

All twelve belong in the construction ERP scope. The depth varies, but none should disappear merely to shorten the menu. Place them inside coherent project workspaces and shared services. A visible record proves that capability exists at that level; it does not prove a complete or reliable end-to-end process.

Area / detail page	Current evidence and status	Need in the new ERP
1. NCR operations Page 13	PARTIAL: populated Non-conformance records, resolution, verifier and timestamps. E56-E59.	Essential from mobilisation: contain defective work, resolve the cause, verify the remedy, and trace rework consequences.
2. Asset management Page 14	PARTIAL: 20 equipment records; crane serial/cost, maintenance links and reliability metrics. E61-E64.	Essential: know what is available, where it is, who has it, whether it is fit for use, and what it costs. Separate owned/hired plant and client handover assets.
3. Payments & cash flow Page 15	PARTIAL / BLOCKED REPORT: posted receipts and bills exist. Company cash report requires a policy; visible policy and cash-forecast lists are empty. E70, E73-E76, E91-E92.	Essential: expected versus actual money, approved obligations, collections, payment scheduling, advances, retention and liquidity.
4. Subcontracting Page 16	SUBSTANTIAL PARTIAL CHAIN: service-entry/certificate/bill links, finance agreement, retention terms and balances. E29, E70-E72.	Essential: manage each package from selection and award through measured work, changes, payment and final settlement.
5. Item cost breakdown Pages 17-19	PARTIAL: BOQ/location report, source facts, WBS budgets and approved cost-to-complete forecasts. Samples remain coarse. E66-E69, E87-E90.	Essential: one explainable item ledger with budget, quantities, resources, incurred cost, commitments, payments and completion forecast.
6. Waste management Page 20	LIMITED: empty scrap register and unsaved stock-scrap form. Full site waste chain not demonstrated. E77-E78.	Essential: material loss and environmental disposal/reuse records; costs and quantities linked to the work that caused them.

Status key: PARTIAL = useful records or controls observed, with workflow depth unverified. LIMITED = narrow or unpopulated evidence. BLOCKED REPORT = a specific report could not produce results under the inspected setup. None means proven absent throughout the system.

Your 12 areas: controls and visibility

Expanded scope map 2 of 2. Proposed requirements below are design inputs, not claims about existing functionality.

Area / detail page	Current evidence and status	Need in the new ERP
7. Variations / change requests Page 21	PARTIAL: 17 variation records; owner/subcontract distinction, pricing, approval and certification states. E85-E86.	Essential: capture an early change event, assess scope/cost/time, obtain authority, and carry the outcome into budgets, orders and programme.
8. Programme & delay Page 22	PARTIAL: accepted progress facts; screen explicitly makes P6 the schedule master. Delay/EOT chain not demonstrated. E79-E80.	Essential: baseline/current plan, look-ahead, constraints, delay events, mitigation and time-impact evidence.
9. KPI management Page 23	PARTIAL: CPI/SPI, project financial/progress metrics, exception activities, turnaround data and generic dashboards. E53-E54, E66, E82-E84.	Essential: definitions, owners, targets, trends, data freshness and corrective actions; a chart alone is insufficient.
10. Keeping records Page 24	PARTIAL: revisions, attachments, diaries, lifecycle rows and source links. E37-E45, E57-E59, E82.	Foundation: authoritative evidence, controlled revisions, search, distribution, retention and export throughout the project.
11. Approval workflow Page 25	PARTIAL: staged certificate review, named forecast approver, quality verifier, document review and activity queues. E15, E54, E59, E71, E82, E90.	Foundation: consistent, policy-based routing with next owner/due date, rejection/resubmission, delegation and version-aware decisions.
12. Tracking / transparency Page 26	PARTIAL: project facts link to source records; dates, provenance and some approver fields exist. E69-E72, E81-E82, E90-E92.	Foundation: show where work stands, why it is blocked, who acts next, and how every total was derived, within access rights.

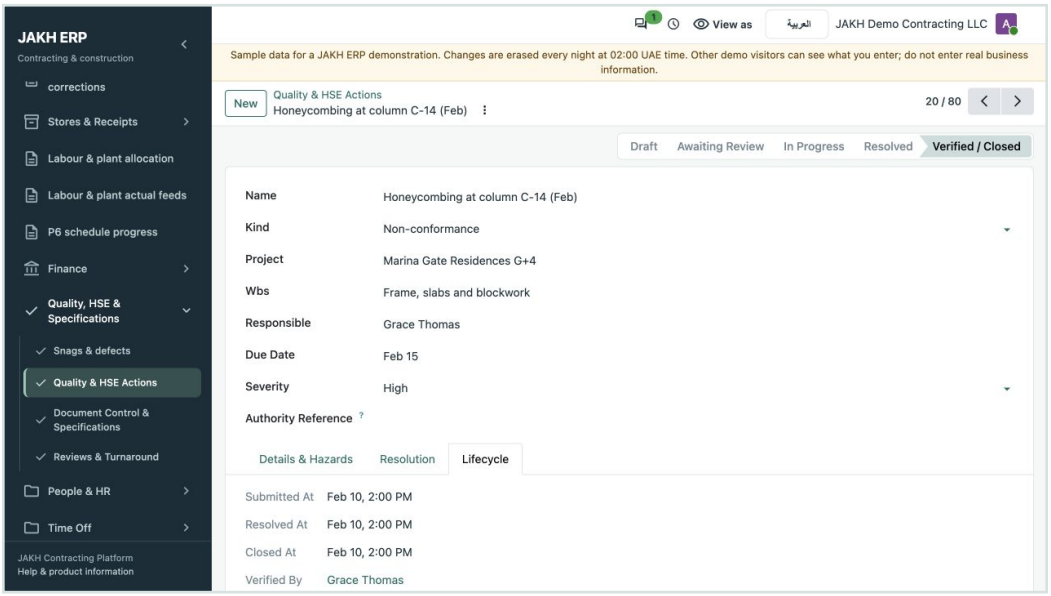
How these fit together

The common spine is company -> project -> building/floor/zone -> work item/package, with linked programme activity, resource, party and evidence. Records, approvals and tracking serve every workflow. Quality, cost and schedule consequences must travel with the originating event.

Example: failed inspection -> NCR -> authorised repair -> labour/material/plant use and waste -> reinspection -> accepted progress. If scope or time changes, link the change/delay assessment. Do not assume every NCR automatically creates a billable variation.

1. NCR operations

Existence: confirmed at record level; completeness: partial. Route: Quality, HSE & Specifications -> Quality & HSE Actions. E56-E59.



E59. A closed honeycombing NCR has project/WBS, responsibility, severity, lifecycle dates and a named verifier. Resolution is a separate free-text section.

What exists; what remains uncertain

The register mixes inspections, non-conformances and risk assessments. The sampled NCR has one attachment and a generic correction statement. Its responsible person and verifier are the same in this sample. Structured requirement breach, root-cause analysis, distinct corrective-action tasks, effectiveness review and linked rework costs were not visible; independence rules were not exercised.

Needed workflow and simple UI

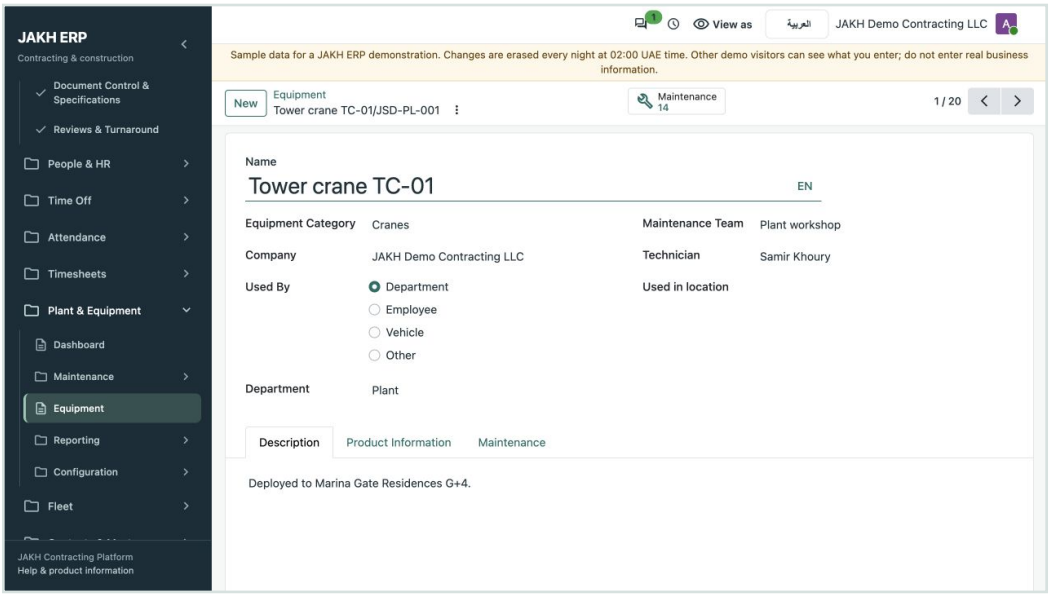
Essential. QA/QC raises the issue against an inspection, specification and exact location; the site team contains it; the responsible contractor proposes a disposition; required technical/client reviewers authorise it; the team implements and supplies evidence; QA/QC verifies and closes or reopens. Record cause, prevention, owners, deadlines and cost/time impacts. Keep it in Project -> Site & Quality, with one case timeline and the action required now.

Test: trace one rejected concrete item through repair, reinspection and linked costs; show why it may be closed. Where policy requires independent verification, prove the resolver cannot provide that approval.

Reference reviewed: the public QIC NCR toolkit description covers a form, procedure, register and dashboard, including cause analysis, corrective action and effectiveness checks. Paid files were not acquired or inspected. This is a process reference, not certification evidence or a copied template.

2. Asset management

Existence: equipment and maintenance confirmed; a complete asset lifecycle is not demonstrated. E61-E64.



E62. Tower crane TC-01 has a maintenance team and 14 linked maintenance records. The project deployment is in description text; Used in location is blank.

Preserve, then connect

Twenty equipment records are visible. The crane has a serial number, cost, vendor/model/warranty fields and reliability measures. Fleet also has navigation entries. These do not establish fixed-asset accounting, hire control, transfer history, inspection/calibration validity, fuel/use reconciliation or client asset handover.

Needed workflow and simple UI

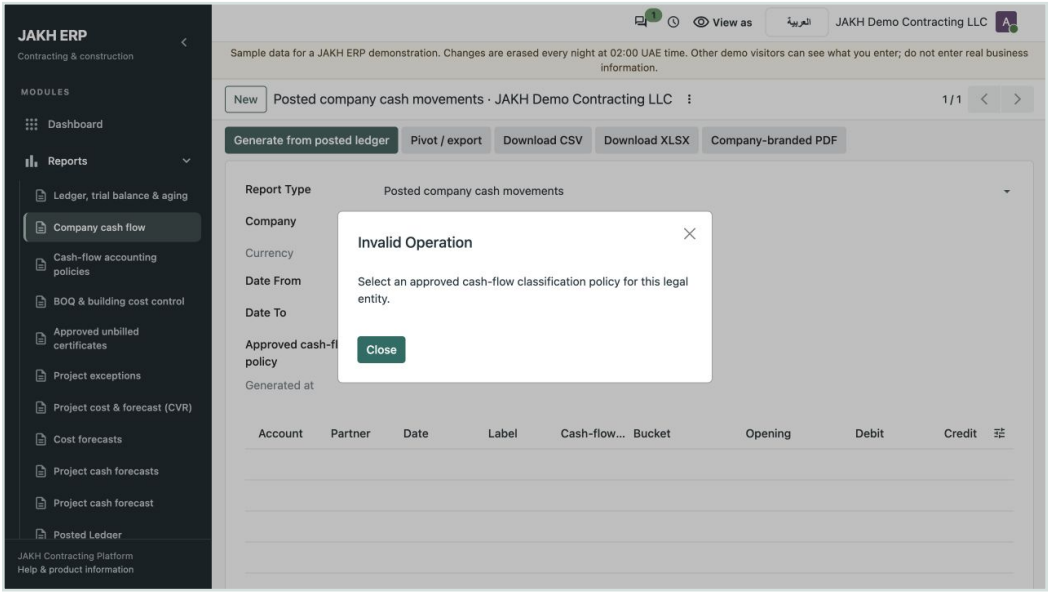
Essential, with three related tracks. Owned plant/tools: acquire -> identify/tag -> assign/transfer -> use -> maintain -> dispose. Hired plant: hire terms -> on-hire -> utilisation/downtime -> off-hire -> invoice check. Installed client assets: install -> test/commission -> warranty/O&M -> handover. Link rather than duplicate asset identities across operations and finance.

The plant manager owns availability and service dates; site custodians own location/use; finance owns valuation and depreciation policy. Link hours, fuel, repairs, hire, depreciation or approved internal charges to the appropriate project/work item. Keep Resources -> Assets as the shared register, and show assigned assets in each project.

Test: transfer a crane between projects, stop scheduling it while out of service, reconcile its utilisation/cost, and find its inspection evidence. Separately retrieve one installed pump's commissioning and warranty pack.

3. Payments & cash flow

Existence: payment evidence confirmed; forecast depth unverified; company cash report blocked by its policy prerequisite. E70, E73-E76, E91-E92.



E74. Company cash generation requires an approved classification policy. E75 shows an empty visible policy register; E76 shows no project cash-forecast records.

What the evidence supports

The building report links to eight posted payment sources totalling AED 15,675,795.22. One source has balanced bank/receivable journal lines and an invoice reference. The report explicitly does not assert bank clearance. A subcontract bill shows AED 257,484 gross work, AED 25,748.40 retention and AED 231,735.60 due. Certification, invoicing, due balances and cash are different facts.

Needed workflow and simple UI

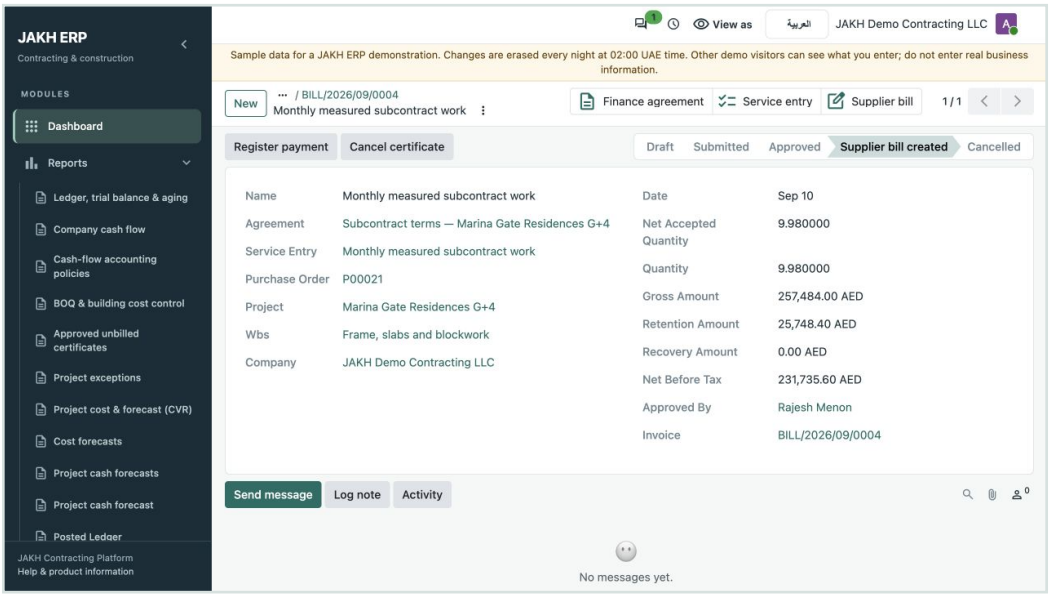
Essential. Finance and QS maintain expected receipts and outflows from payment terms, certificates, invoices, payroll, orders, taxes, advances and retention. Use a rolling weekly view for near-term liquidity and monthly project-to-completion outlook, with base/late-payment scenarios. Reconcile actual bank movements, forecast dates and invoice allocations; explain unmatched or overdue items.

Show project cash in Project -> Commercial, and the same records in Finance for company-wide scheduling and reconciliation. Display planned, approved, due, paid/received and bank-reconciled states separately. Include approval and bank authority checks; report access must not imply payment authority.

Test: follow partial settlement, advance recovery, retention release and a late receipt through both project and company views. Validate the cash-policy setup before relying on the company report. No payments were initiated in this study.

4. Subcontracting management

Existence: a substantial measurement-to-bill chain is visible. Award, exceptions and settlement remain to be validated. E29, E70-E72.



E71. The supplier certificate connects service entry, PO, project/WBS, finance agreement, approver, retention and bill.

Current strengths and gaps

The linked agreement holds retention percentage/cap, advance/recovery terms, balances and eight certificates. The bill links back to the service entry, PO and supplier certificate. Separate application, amendment and correction menus exist. Tender comparison, scope boundaries, securities, insurance expiry, disputed measurements, backcharges and final settlement were not walked end to end.

Needed workflow and simple UI

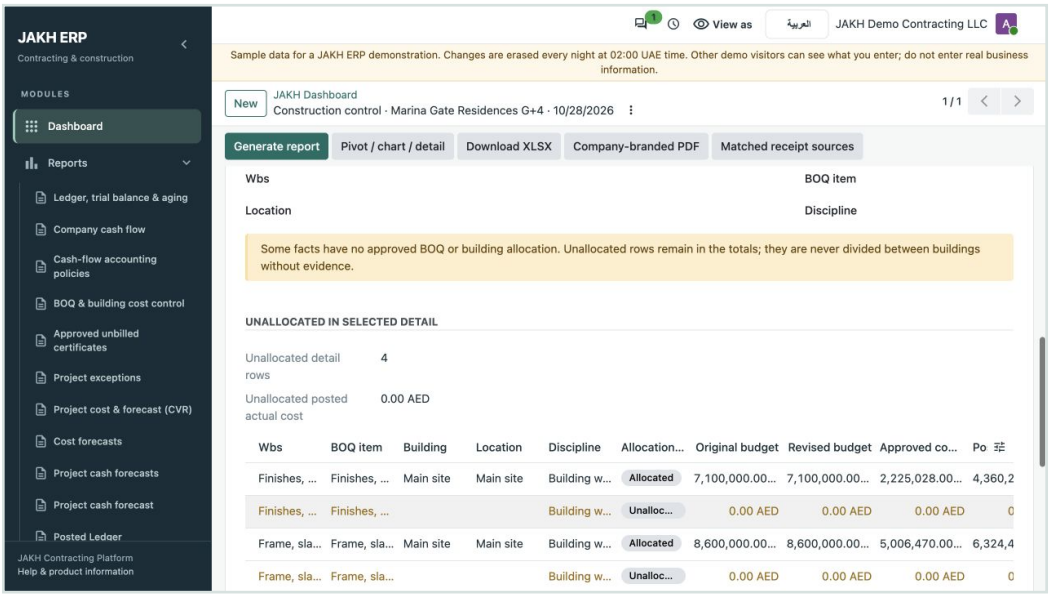
Essential. Procurement/QS prequalify and compare bids -> approve package scope and award -> record mobilisation obligations -> receive the subcontractor's application -> measure/inspect accepted work -> assess deductions and changes -> approve certificate -> match bill -> pay -> settle and release retention against agreed conditions.

Use one package workspace: scope/BOQ, commitments, progress, changes, quality obligations and money. Keep claimed, assessed, certified, invoiced and paid amounts distinct. Link NCRs, rectification, securities and handover readiness. Buying provides cross-project queues; project and portal views reference the same authorised records.

Test: certify only accepted quantities from a partly rejected application, amend scope, apply an evidenced deduction, then reconcile the package account without counting both the service receipt and bill as separate incurred cost.

5. Cost analysis for each site item

Existence: BOQ/location cost reporting and source lineage are confirmed. Required tower/resource granularity is not yet demonstrated. E66-E69, E87-E90.



E67. The report distinguishes original/revised budgets, commitments and posted cost, with BOQ/building/location fields and explicit unallocated rows.

A useful foundation with a granularity gap

The sampled building uses four broad WBS/BOQ groups, generally allocated to Main site. This does not demonstrate the cost of a particular floor slab, door type or installation activity. Source facts distinguish commitments, received exposure, posted cost and revenue/billing. Approved forecasts include actual-cost snapshots and remaining costs; report rules avoid silently dividing WBS forecasts among buildings.

One WBS form shows Forecast Amount = 0 while its dated approved forecast is AED 8,428,000 (E88-E90). The report says it uses the approved forecast. The UI should identify that authoritative value and explain the other field; the difference is not proof the report calculation is wrong.

Required grain and ownership

Essential. QS/cost control defines work items and cost codes; planning supplies quantities and dates; site/stores supply consumption and progress; procurement supplies commitments; finance supplies postings/payments. Analyse by company, project, location, work item/WBS, resource type and period. Map client BOQ lines to internal work items with explicit, versioned allocations; they are not necessarily one-to-one.

Simple UI: select an item and location once, then see Budget / Incurred / Paid / Forecast with expandable resource and source detail. Never label all four as Actual. Pages 18-19 define the measures and a reconciliation example.

Cost dictionary: compare like with like

Proposed definitions for agreement with cost control and finance. These are requirements, not a claim that every measure is implemented.

Measure	Meaning and control
Original budget	Frozen approved cost plan: quantities x resource rates, plus explicitly allocated indirect cost and risk. Keep the original version.
Revised budget	Original budget plus approved budget changes/transfers. An owner selling-price variation must not automatically become an equal cost-budget change.
Plan to date / earned value	Planned value is budgeted work scheduled by the status date. Earned value is budgeted value of accepted work completed. Both must use the same approved baseline and measurement rules.
Commitments	Approved purchase/subcontract obligations, including controlled amendments. Show total and remaining uninvoiced/unperformed exposure separately; total commitments are not extra cost to add to actuals.
Incurred actual cost	Posted cost plus approved period-end accruals, net of corrections and credits, counted once. Stock receipt, issue, installation and invoice may represent different stages, not four costs.
Cash spent / received	Allocated payment or receipt amounts and dates. Separate advances, taxes and retention; show reconciliation status. Cash spent may precede or follow the related incurred cost.
Cost to complete	Reviewed remaining committed work + remaining uncommitted work + explicit risk allowance. Identify assumptions, rates, productivity, owner and update date.
Forecast final cost / EAC	Incurred cost + cost to complete. Show variance to revised budget; a favourable forecast is not realised profit. Keep prior approved forecasts.
Revenue / margin	Separate contract value, approved/pending changes, claimed, certified, invoiced and collected amounts. Compare compatible revenue/cost forecasts for margin; cash balance is a separate measure.

Every number needs a basis

Display cutoff date, company/currency, tax basis, unit of measure, baseline/revision, allocation status and data freshness. For each cost type expose budget quantity/rate, actual quantity/rate, productivity, waste and the source documents. Use material, labour, plant, subcontract and indirect-cost categories.

Never hide missing allocations, stale forecasts or unreconciled differences. Shared costs require an approved allocation basis and version history. Accruals must reverse or clear when the underlying bill arrives. A business owner must approve the precise posting and valuation rules.

One item, four different answers

Illustrative acceptance fixture only - invented figures, not extracted from JAKH. Example: one floor slab work item, one cutoff date, AED, tax excluded.

Question	Example answer / explanation
What was planned?	Original budget 100,000; approved cost-budget change +10,000; revised budget **110,000**.
What has it cost so far?	Posted cost 52,000 + approved unbilled accrual 8,000 = incurred cost **60,000**.
What is committed?	Approved package/order obligations total 80,000. Of these, 40,000 is already included in incurred cost; 40,000 remains. Other incurred labour/plant costs are 20,000.
What cash has left?	Payments **45,000**: 35,000 settling bills plus 10,000 advance. This does not turn incurred cost into 45,000 or 105,000.
What will remaining work cost?	Remaining committed work 40,000 + other remaining work 5,000 + risk 3,000 = cost to complete **48,000**. Future cash timing also reflects unpaid incurred costs and advance recovery.
Where will it finish?	Incurred 60,000 + remaining 48,000 = forecast final cost **108,000**. Revised budget 110,000 - forecast 108,000 = forecast headroom **2,000**.

Break the same item into resources

Expand the slab into concrete, reinforcement, formwork, labour, plant and allocated site overhead. Each has planned quantity/rate, actual consumed quantity, accepted installed quantity, loss/rework, incurred cost and remaining forecast. Use a common location and cutoff; aggregate the detail back to the item, floor and project.

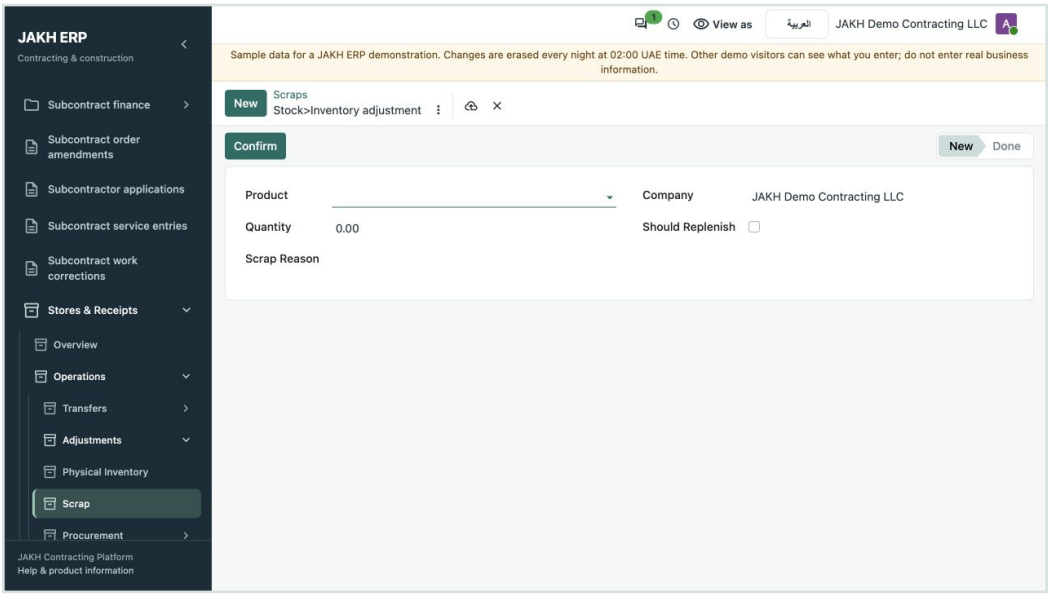
Required reconciliation tests

1. Every item total equals its contributing facts, with no source counted twice.
2. Cost posted without an item remains visibly unallocated until an authorised allocation.
3. A later bill replaces/clears the accrual rather than adding a duplicate cost.
4. Paid amounts trace to invoice/advance allocations and bank reconciliation.
5. Changes preserve the original budget and prior approved forecasts.

For quantity control, reconcile opening usable stock + receipts/transfers in against installed or otherwise consumed quantity, returns/transfers out, closing usable stock and recorded loss. Define measurement units and avoid counting waste again if it is already included in a consumption category.

6. Waste management

Existence: a stock-scrap function is visible; full construction waste control is not demonstrated. E77-E78.



E78. Unsaved scrap form: product, quantity/unit, reason, company and replenishment. It was discarded without entry or confirmation.

Two needs must be covered

Material efficiency: distinguish allowed waste, avoidable loss, damage, theft, rejected work, recoverable offcuts and rework. Link issued/returned/installed quantities to the item, location and responsible trade. An inventory adjustment alone does not explain what happened on site.

Site waste handling: classify waste, segregate/store it, arrange collection, record weight/volume, carrier and destination, then attach acceptance/weighbridge/disposal evidence. Track reuse, recycling, recovery proceeds and disposal cost. Confirm jurisdiction and contract requirements before specifying controlled waste paperwork.

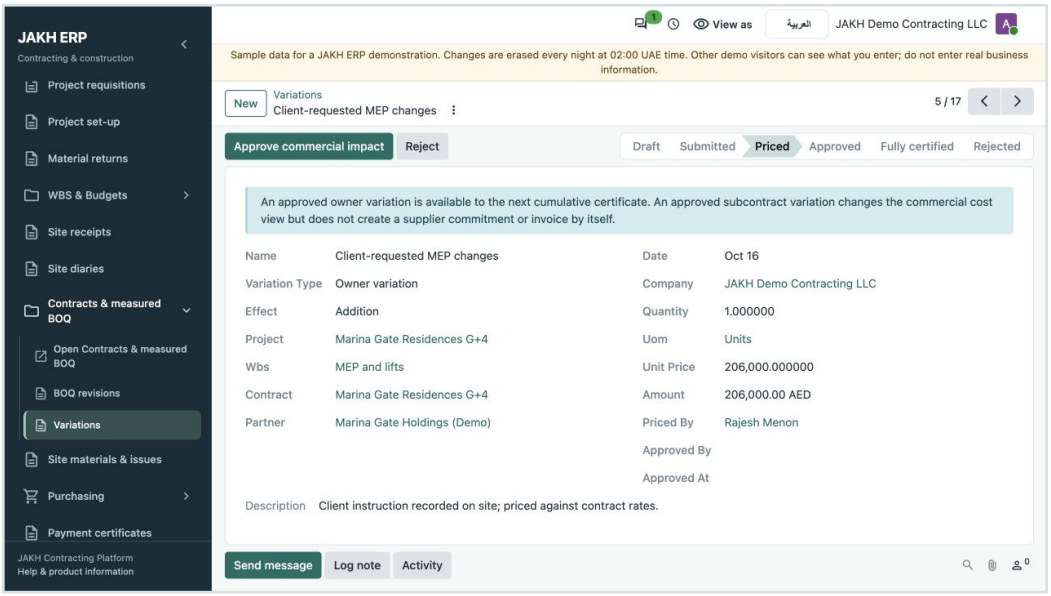
Needed workflow and simple UI

Site/store staff record waste against work or an NCR; the site/environmental owner selects the handling route; an authorised person approves the write-off or disposal; receipt/evidence closes the movement; cost control allocates the impact. One Project -> Site & Quality waste register should connect physical movements and cost, with stock transactions remaining controlled linked records.

Test: reconcile reinforcement received, used, returned, reusable offcuts and lost/scrapped quantity for one floor; follow one disposal load to acceptance evidence and its cost. Define waste-rate denominators by material and units; do not combine tonnes and cubic metres in one total.

7. Variations and change requests

Existence: commercial variation records confirmed; the upstream change-event and time-impact chain remains partial. E85-E86.



E86. A priced owner variation links project, WBS and contract. The instruction is mentioned in free text; structured notice/time-impact fields are not visible.

What exists

The register includes Draft/Submitted/Priced/Approved/Fully certified and Rejected stages. The UI explains that an approved owner variation feeds certification, while a subcontract variation does not itself create a commitment or invoice. Preserve that distinction. Internal design changes, early warnings and potential changes were not demonstrated as a separate connected intake.

Needed workflow and simple UI

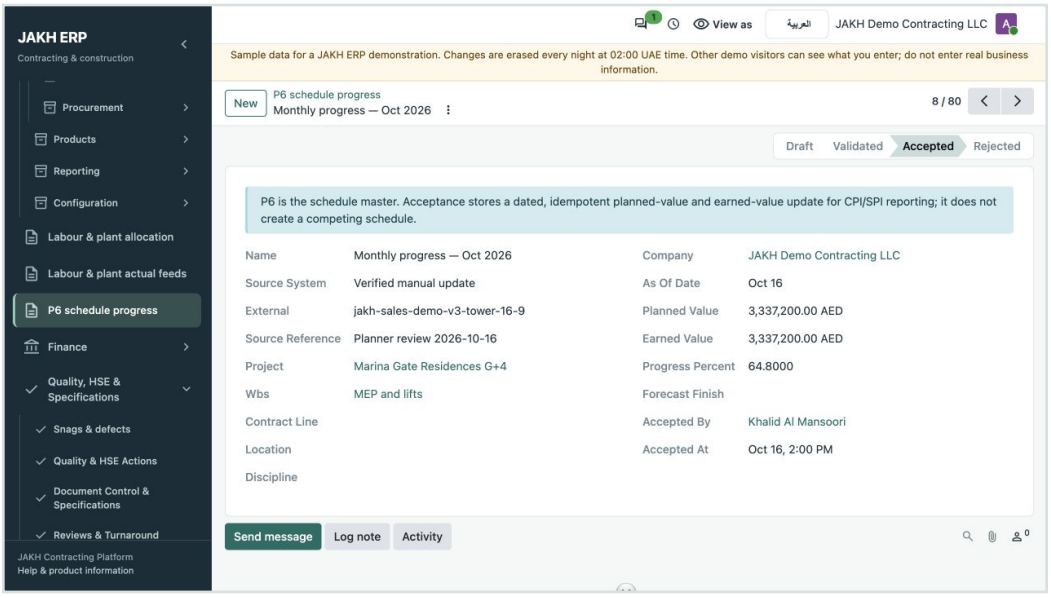
Essential. Capture a change event from a client instruction, RFI, revised drawing or site condition. Identify affected scope/location, initiator, responsibility and contract references. Track notice requirements, cost/time assessment, quotations, review, decision and implementation. Keep potential, instructed, submitted, approved, disputed, rejected and withdrawn outcomes understandable.

QS/PM own assessment; planning owns time effects; the designated internal/client authority owns decisions. Link approved outcomes to budget revision, contract/BOQ, subcontract/PO amendment, programme revision, forecasts and certificates. Internal approval and client acceptance are separate decisions. Configure notice dates from the actual contract; do not hard-code a universal deadline.

Test: an urgent instructed change can proceed under recorded authority while its price is disputed; costs, notice dates and programme effects remain visible. Use one Changes workspace with an evidence timeline, not repeated data entry in each downstream record.

8. Programme & delay

Existence: progress ingestion and acceptance confirmed; full planning and delay management not demonstrated. E79-E80.



E80. The screen says P6 is the schedule master. This accepted manual update has PV/EV and progress; forecast finish and location are blank. No activity-level link is visible.

Separate progress reporting from controlling time

A percentage or SPI does not explain the critical activity, lost days or recovery plan. The inspected record is a dated WBS progress fact. It does not demonstrate dependencies, calendars, baseline comparison, critical path, float, resource loading, look-ahead constraints or an EOT assessment.

Needed workflow and simple UI

Essential. Planner owns an approved baseline and dated updates. Site teams report actual starts/finishes and remaining work, linked to accepted evidence. Surface a near-term look-ahead with prerequisites such as design release, access, materials, inspections, labour and plant. Record delay events with discovery/date range, affected activities, cause, evidence, responsibility, mitigation and review status.

Connect delay assessments to notices, change events, the programme version analysed, forecast finish and any time/cost claim. Keep requested, assessed and granted extension distinct. A planner/contract administrator must validate analysis and contractual interpretation; a metric alone must not decide entitlement.

Architecture decision: retain a clearly defined P6 exchange or build original scheduling capability, according to operational need. Either requires one authoritative programme and traceable versions. **Test:** a late approved drawing blocks a floor activity; show the dependency, mitigation, notice and updated finish without overwriting the baseline.

9. KPI management

Existence: metrics and dashboards confirmed; a complete KPI definition/ownership/action cycle is unverified. E53-E54, E66, E82-E84.

Construction dashboards already expose CPI/SPI and financial/physical progress. Saved Project dashboards emphasise tasks, logged hours and task closure, while document turnaround has reviewer/due-date data. These are useful components, but a KPI needs an agreed meaning and accountable response.

Recommended starting KPI	Definition / owner / response
Cost performance	CPI = earned value / compatible incurred actual cost, where cost > 0. Cost controller investigates adverse cost/productivity drivers; show excluded/unallocated costs.
Schedule performance	SPI = earned value / planned value, where plan > 0; planner also tracks milestone slippage and critical constraints. SPI is not a count of delay days.
Completion forecast	Revised budget minus forecast final cost, plus forecast coverage/freshness. QS owns the cost-to-complete review and corrective action.
Quality	Open/overdue NCR count, closure duration, recurrence and rework cost. QA/QC owns review; normalise comparisons by work/inspection volume where appropriate.
Buying / subcontract	Required-on-site delivery reliability and overdue package actions. Procurement/PM owns expediting and recovery.
Cash / commercial	Overdue receivables/payables, certified-but-unbilled value and forecast cash shortfall. Finance/QS own collections, billing and payment planning.
Resources / waste	Productive asset hours / available planned hours; material waste / defined input quantity. Plant/site owners investigate downtime and loss.
Approvals / records	Overdue reviews and turnaround by document type/owner; current approved document coverage. Document control and workflow owners resolve bottlenecks.

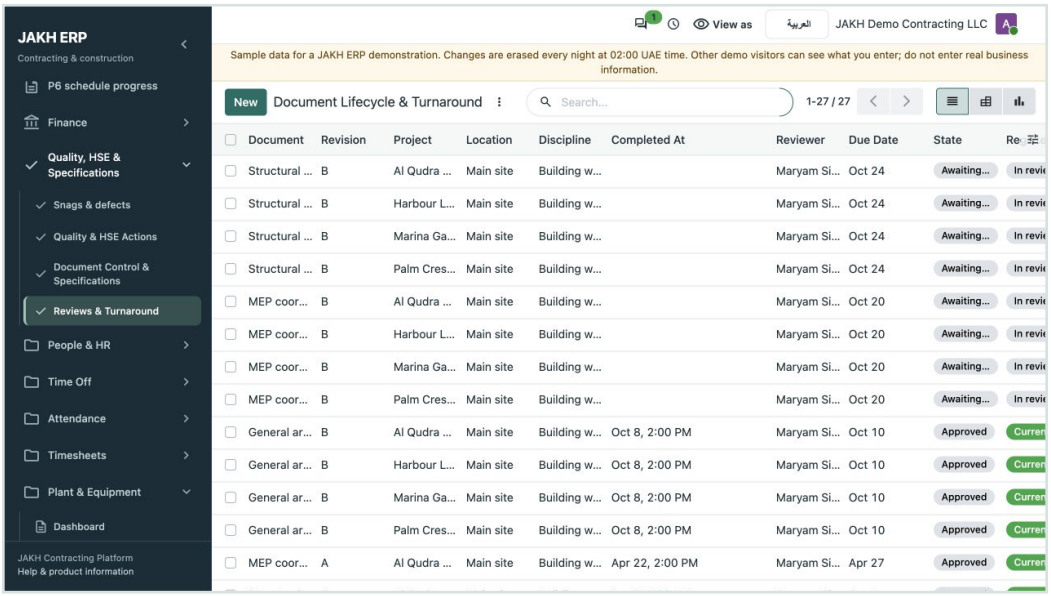
One metric definition, several contextual views

Store formula/version, unit, population, exclusions, period, baseline, target, threshold, owner, refresh time and source link. Support trend and explanation, then assign an action with due date. Start each role with a few actionable measures; let users expand the supporting detail.

Test: a dashboard total reconciles to its records under the same filters; zero, missing and not-applicable values differ. Crossing a threshold creates one owned action and stops duplicate reminders. Targets need business agreement; this study does not invent performance benchmarks.

10. Keeping records

Existence: document revisions, attachments, site diaries and review records confirmed; retention and distribution controls remain unverified. E37-E45, E57-E59, E82.



E82. Review records expose document/revision, project, reviewer, due date, completion and status; age/cycle measures are available in the wider table.

The record must explain what was known then

Preserve a document ID and revision, issue purpose, creator/source, project/location/package, timestamps, review decision and linked work. The current approved drawing should be clear alongside history. Approval and issue-for-construction are separate meanings and must be labelled consistently; the register exposes both fields.

Needed workflow and simple UI

Foundation across all modules. Capture -> classify -> review/revise -> approve -> issue/distribute -> acknowledge where required -> supersede/archive. Link correspondence, photos, inspections/tests, daily reports, instructions, notices, measurements, meetings, commercial evidence and handover packs. A chatter attachment is not by itself a controlled document register.

Document control owns numbering/revision/distribution; originating teams own content; project/finance owners set retention and access rules. Search by project, floor, work item, party, date and record type. Define offline photo capture/sync, permissions, version conflicts, backups, export and retention/hold behaviour in engineering validation.

Test: reconstruct which approved revision and instruction supported a particular pour/certificate on a historical date, retrieve its evidence, and export an indexed package without losing relationships. Show relevant records inside each workflow; keep one authoritative copy.

11. Approval workflow

Existence: several document-specific routes and approver histories are visible; universal queue coverage and enforcement remain untested. E15, E54, E59, E71, E82, E90.

Foundation. Use one approval service and personal queue, with distinct policies for each business object. The table below proposes ownership for validation; it is not a statement of the current permission configuration.

Object / trigger	Proposed responsibility and decision boundary
Tender / budget / forecast	Estimator or QS prepares; PM/commercial authority reviews; delegated authority approves within value/risk limits. Preserve priced and approved versions.
Purchase / subcontract award	Requester -> technical/budget check -> procurement/commercial -> authorised award. Receiving and paying require separate checks.
Inspection / NCR / snag	Site/contractor submits -> QA/QC/technical review -> required consultant/client sign-off -> verified closure. Configure independence and hold-point rules.
Change / time assessment	Originator -> QS and planner assessment -> PM/internal authority -> client/consultant decision where needed. Keep commercial and time decisions traceable.
Measurement / certificate	Site validates quantity -> QS assesses -> commercial/finance checks -> authorised certificate. Supplier claim, certificate, bill and payment remain distinct.
Bill / payment / release	Match supporting obligation and evidence -> finance review -> authorised payment/release -> bank reconciliation. No dashboard permission grants bank authority.
Document / handover	Author -> discipline reviewers -> approval -> authorised release. Handover also checks technical, client acceptance and commercial obligations.

Controls and UI that every route shares

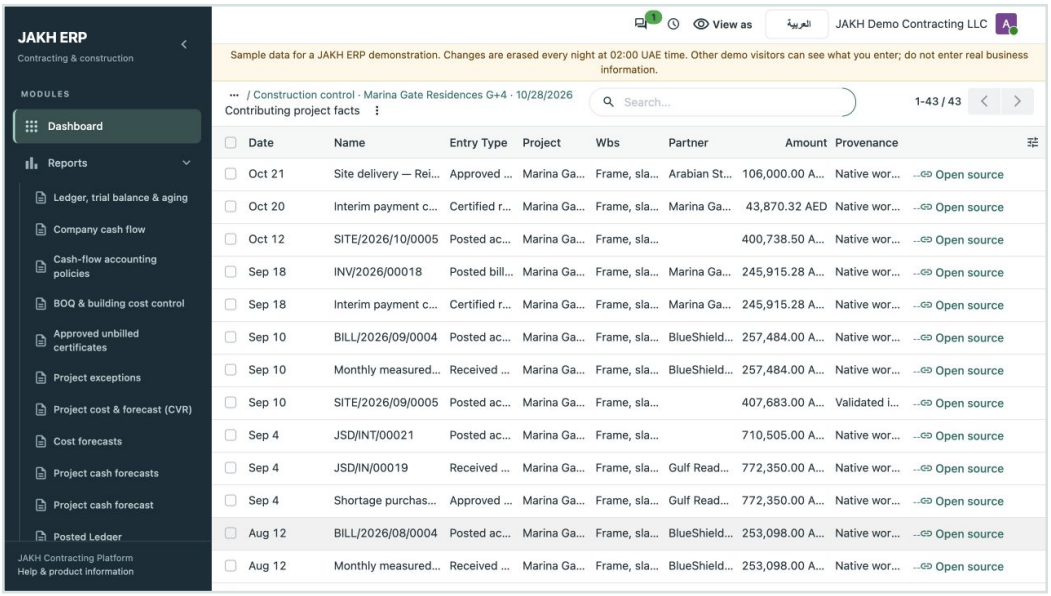
Show current step, assigned person/role, due date, blockers and decision history on the record and in My Work. Support sequential/parallel review, amount/risk limits, substitutes, absence delegation, escalation, rejection reasons and resubmission. A material edit must invalidate or reroute the affected approval version.

Record policy version, actor, authority, timestamp, comments and document snapshot. Enforce permissions on the server and prevent duplicate/concurrent decisions. Distinguish an operational activity marked Done from a formal business approval.

Test: absence delegation, out-of-limit approval, prohibited self-approval, changed documents and simultaneous clicks all produce correct, explainable outcomes. Queue counts reconcile to pending business records. Read-only inspection did not execute these tests.

12. Tracking and transparency

Existence: source drill-through and provenance confirmed; complete event history and permission boundaries require validation. E69-E72, E81-E82, E90-E92.



E69. Contributing facts show dates, classifications, project/WBS, amounts, provenance and Open source links. One was followed to a supplier bill and certificate.

Keep the good traceability, make it easier to read

The project-control audit lists financial/control facts, including their provenance. This is useful evidence lineage; it is not proof of an immutable who-changed-what audit log. Approver/date fields and chatter support some history, but universal old/new values, deletions, permission changes and restoration were not tested.

Needed behaviour and simple UI

Foundation. Every record answers: what is this, which work does it affect, what changed, who owns the next step, when is it due, and why is it blocked? Use a consistent timeline for submission, revision, decision, execution, posting and settlement. Link each event to its actor, authority, source version and resulting records.

A project overview should summarise exceptions and next actions; users expand only what needs explanation. Every aggregate drills to the same filtered records and shows refresh/cutoff and missing data. Client/subcontractor views show their permitted scope, with confidential internal prices and personal information controlled by policy.

Test: trace a forecast variance to an item, bill/service entry and originating scope change; reconstruct its approval version. An authorised user can export that chain, and an unauthorised user cannot gain access through a deep link. Records/approvals/tracking share one history service rather than three duplicate modules.

A simpler structure to validate

Proposed information architecture only. No visual direction, interface prototype or application code has been produced.

Test a small set of stable work areas, filtered by responsibility. The exact count and names should follow task testing rather than become an arbitrary restriction.

Candidate area	Purpose
My Work	Approvals, assigned work, blockers and overdue actions across permitted projects. Each item identifies project, owner, due date and next action.
Tenders	Opportunities, tender documents, BOQ, estimates, comparisons, revisions and the award handoff.
Projects	Overview, programme/delay, site delivery and quality (NCRs/waste), changes/commercial, and handover, revealed by stage and role. Records and item-cost detail stay in context.
Buying	Cross-project purchasing/subcontract queues and supplier work. Project views filter the same underlying records; they do not create copies.
Finance	Project commercial-to-accounting handoffs, receivables/payables, cash, ledger and period work, with appropriate authority.
Resources	People, assets/plant, stores/catalogues and shared records. Project views show the same assigned resources and their usage/cost.
Administration - utility access	Company setup, permissions, workflow policies and integrations, outside the daily operational menu.

Rules for keeping it simple

Keep project / package / location visible. Show the few actions valid now. Use one canonical record with contextual views. Enter shared facts once; preserve approved baselines and document snapshots with traceable revisions. Put reports where the related work happens.

Repeated BOQ views can represent estimate, awarded baseline and certified measurement - different business meanings. Remove unnecessary re-entry and competing navigation; do not erase the history or controls that make those stages trustworthy.

The twelve requested capabilities do not need twelve more sidebar entries. Records, approvals, tracking and metric definitions are shared services; project and cross-project views link to the same canonical records. Use a persistent project/location selector, a short stage-aware overview, and expandable detail rather than repeating all tabs at each step.

Define an original application

Rebuild boundary and migration study. The current product supplies observable business requirements, not implementation code.

No Odoo application-code dependency

Build original source, data models, APIs, approval/state rules and UI. Do not copy, translate or adapt Odoo modules or application code. No existing source code was inspected during this study. Selection of framework, database and deployment architecture remains a later engineering decision.

Core domain relationships to specify

Company and authority -> party/person -> opportunity/tender -> estimate revision -> contract and BOQ baseline -> project, location and WBS/package -> demand and commitment -> delivery/service/progress evidence -> variation/certificate -> invoice/payment -> handover/defect obligation.

Keep document revisions, financial postings, approval events and operational evidence linked but distinct. Define stable identifiers, permissions, immutable approved snapshots, corrections, search, attachments and audit history across those relationships.

Explicit domain scope also includes quality/NCR and corrective action, asset custody/maintenance/hire, waste movements, change events/notices, programme/delay versions, item/resource cost allocations, cash forecasts and KPI definitions. Shared records, approvals and event history connect these domains.

Finance and supporting ERP scope

Specify original accounting/posting and reconciliation requirements with the finance owner. Inventory the required HR, attendance, leave, timesheet, plant, fleet, master-data and administration functions. Integration decisions require explicit scope; they must not quietly replace requested functionality.

Migration and cutover requirements

Step	Evidence required before production use
Inventory and export	Authorised exports of records, identifiers, attachments, companies, currencies, statuses, approvals and ownership.
Map and trial-migrate	A field/state map and repeatable trial import; preserve relationships and approved historical meaning.
Reconcile	Counts, opening balances, inventory, open orders, project budgets/costs, certificates, claims, retention and attachments checked by owners.
Pilot and cut over	One representative project plus exceptions; parallel reconciliation, backups, rollback and agreed transition responsibilities.

Validate before detailed design

The read-only study is ready for review. Detailed design and application coding remain behind the requested review point.

Priority sequence - the full scope remains

Priority	Outcome
First: navigation and context	Agree roles, canonical records, project/floor/package context, terminology and an understandable work queue.
Next: tower workflow depth	Resolve engineering-to-site, instruction-to-variation/time, package-to-measurement, and technical-to-commercial closeout handoffs.
Then: coherent original ERP	Design and build the lifecycle in increments, including original finance controls and the required supporting modules.
Before replacement	Trial migration, representative-user validation, accounting/control review, security/access testing, integration checks and a controlled pilot.

Tasks for the next validation round

1. Find the approved drawing for one tower floor and confirm which revision may be used.
2. Request a material, follow a partial/rejected delivery, and find the installed quantity.
3. Trace a subcontract package from measured work through payment.
4. Record an instruction, identify cost/time consequences, and follow its approval or dispute.
5. Explain claimed, certified, invoiced and collected values for one period.
6. Find all snags in one unit, verify the fix, and check partial-handover readiness.

Use you and representative estimator, site, procurement, QS/PM, quality/document-control and finance users where available. Measure unassisted completion, time to the correct information, wrong turns, repeated entry and missed handoffs. Set numerical targets after obtaining a real baseline; no staff usability test has yet been run.

Decisions still needed

Confirm actual roles/approval authority, contract types, floor/unit structure, consultant/client/subcontractor participation, document and programme tools, accounting/localisation needs, languages/mobile/offline needs, project volumes and migration depth. A delivery estimate should follow this scope, not precede it.

Terms used in the study

BOQ: bill of quantities. WBS: work breakdown structure. QS: quantity surveyor. RFI: request for information. ITP: inspection/test plan. EOT: extension of time. DLP: defect-liability period. O&M: operation and maintenance.

Twelve validation scenarios

Proposed acceptance work after study review. These scenarios have not been executed against the current ERP.

Area	Evidence needed to accept the workflow
1. NCR	A failed inspection reaches authorised repair and verified closure; cause, effectiveness, location and rework costs remain linked.
2. Assets	One owned and one hired asset can be assigned, moved, maintained and costed; one installed asset has an exportable handover pack.
3. Cash	Partial payments, advance recovery, retention and late receipts reconcile between project forecasts, ledgers and bank evidence.
4. Subcontracting	A partly rejected claim produces the right certificate, deduction, bill and payment; package totals and final retention reconcile.
5. Item cost	One floor item reconciles original/revised budget, resources, incurred cost/accruals, remaining obligations, cash and forecast without duplication.
6. Waste	One material quantity balance closes; a disposal/recovery movement has ownership, destination evidence and allocated cost.
7. Changes	One instructed but disputed change retains notices, assessment, decision versions and downstream budget/order/programme links.
8. Programme / delay	One delayed prerequisite is traced to affected activities, mitigation, forecast finish and a versioned time assessment.
9. KPIs	A metric reconciles to its source population and produces one accountable action when its agreed threshold is crossed.
10. Records	A historical work event can be reconstructed using the correct document revisions, correspondence, photos and approval evidence.
11. Approvals	Test delegation, limits, rejection, changed documents and concurrency; the personal queue matches pending business decisions.
12. Transparency	A permitted user traces a dashboard total to source events; a user outside that scope cannot access it through links or exports.

Sequence and release priorities

First: agree IDs, location/work-item structure, definitions, records, approvals and history. **Then:** validate integrated project scenarios across all twelve areas; deepen NCR/waste, programme/delay and cash/item allocation where evidence is weakest. **Before release:** prove roles, reconciliation, migrations and recovery with representative users.

All twelve remain in scope. Advanced analytics and automation can follow reliable operational capture; essential site, cash, quality and approval controls must not be postponed merely for a cleaner-looking first screen.

Evidence and limits

Primary source: live UI at https://demo.jakh.net/ on 28 October 2026. Evidence IDs match saved screenshot and accessibility-snapshot prefixes.

Evidence	Observed route / role / purpose
E02, E39, E41	Administrator dashboards and complete expanded sidebar. E41 is an accessibility inventory, not a single screenshot.
E05-E07	My Company / USD: estimate EST/2026/00004 -> linked project and project shortcuts.
E15-E17	My Company / USD: measured client certificate -> contract -> cash/closeout tab.
E19-E22	My Company / USD: material-shortage requisition -> P00005 -> JD2/IN/00001 receipt.
E28-E29	My Company / USD: subcontract service entries and submitted review record.
E30-E38	Site Engineer, JAKH Demo Contracting LLC / AED: project portfolio, G+4 project, snags and diary.
E40, E42	Administrator, fuller company: G+4 construction report and schedule-progress register.
E43-E45	Administrator, fuller company: document register, G+4 drawing revision history and category menu.
E46-E49	Administrator, fuller company: variations and the approved completed-project final account.
E50-E52	Estimator: role dashboard, CRM entry and eight BOQ studies.
E53-E54	Project Manager: role dashboard and two-item activities/approvals queue.
E55	Administrator returned to original My Company context and normal browser viewport.

Scope limits that affect the conclusions

The first administrator view selected My Company (USD); role previews opened JAKH Demo Contracting LLC (AED), which has fuller samples. Empty initial lists are not proof of absent features. The study inspected a G+4 building and villa/fit-out examples, not high-rise operating scale.

Only navigation, view filters, role previews, record/form reading and report generation/viewing were used. Unsaved scrap and report criteria were discarded. Generated report-view records are the only new records from these inspections. No business document was edited, approved, posted, paid, closed or deleted; no source-code audit, migration, reconciliation or state-transition test was performed.

Accessibility observations are risks to verify: truncation, small/dense text, colour-dependent cues and tab-state semantics. Keyboard/focus, screen-reader behaviour, measured contrast and error recovery were not tested. Screenshot blank capture margins are omitted only in PDF placement; visible content is unchanged. Early clipped capture E07 is excluded from visual conclusions in favour of E33.

Follow-up evidence and reference

Administrator in JAKH Demo Contracting LLC / AED, then restored to My Company. Public reference reviewed separately; no ERP data was sent to it.

Evidence	Observed purpose / result
E56-E59	Quality register -> building Non-conformance -> Resolution -> Lifecycle. Populated, closed case; no transition executed.
E60	Public QIC NCR toolkit product description, accessed 28 October 2026. Paid documents not inspected.
E61-E64	Equipment register -> tower crane -> Product Information -> Maintenance. Identity, cost, location gap and reliability fields.
E65-E69	BOQ report selector -> dashboard-generated building report -> detail/optional columns -> source facts. Selector did not offer the existing project; dashboard route succeeded.
E70-E72	Source fact -> subcontract bill -> supplier certificate -> finance agreement. Quantities, deductions, approvals and balances.
E73-E76	Company cash report -> policy prerequisite error -> empty policy register -> empty project cash-forecast register.
E77-E78	Empty scrap register -> unsaved form, then discard. No scrap or inventory adjustment created.
E79-E80	Schedule register -> accepted building progress record. P6 named as master; manual source shown.
E81-E84	Project-control facts, document turnaround and saved generic dashboards. Record lineage differs from full change-event auditing.
E85-E86	Variation register -> priced building MEP change. Commercial lifecycle and scope links observed.
E87-E90	WBS -> budget record -> forecast register -> approved structural forecast. Authoritative dated forecast differs from simple WBS field.
E91-E93	Matched receipt sources -> posted journal entry -> original dashboard/company restored. No payment/reversal action executed.

Source and interpretation limits

QIC Management Systems: NCR Management Toolkit public description. Its form/procedure/register/dashboard outline informed the NCR comparison. The requirements and UI placement in this report are our proposed construction brief; they do not reproduce or validate the paid toolkit or establish ISO compliance.

An empty list applies only to the inspected role/company/filter context. Fields or capabilities not visible in sampled records are labelled unverified, not globally absent. Sample totals and same-person approvals are evidence of these examples, not proof of accounting correctness or permission enforcement. The public demo can change or reset. Detailed design and coding remain subject to your study review.